# Bihar Engineering University, Patna

## B.Tech 4th Semester Examination, 2024

*Course : B.Tech*
*Code : 101406*

*Subject: Materials, Testing & Evaluation*

*Time: 03 Hours*
*Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are NINE questions in this paper.*
*(iii) Attempt FIVE questions in all.*
*(iv) Question No. 1 is compulsory.*

***Q.1 Choose the correct option / answer the following (Any seven question only):-*** **[2 x 7 = 14]**

(a) Slump test for concrete is carried out to determine
(i) Strength
(ii) Durability
(iii) Workability
(iv) Water content

(b) Which of the following is an example of a thermoplastic?
(i) Melamine
(ii) Urethane
(iii) Epoxide
(iv) Acetal

(c) Consider the following statements regarding the advantages of plywood:
A. It has good strength both along as well as across the grains.
B. It will not shrink or swell across the grains.
C. It can be curved into desired shapes.
Which of the above statements is/are correct?
(i) A and C
(ii) A and B
(iii) B and C
(iv) B only

(d) A cylindrical bar of 25 mm diameter and 1.2 m length is subjected to a longitudinal strain during a tensile test is four times the lateral strain, what is the value of shear modulus? (Take the modulus of elasticity as $1.2 \times 10^5$ N/mm$^2$)
(i) 5.8 kN/mm$^2$
(ii) 4.8 kN/mm$^2$
(iii) 58 kN/mm$^2$
(iv) 48 kN/mm$^2$

(e) Which one of the following statements is not correct for Aluminium?
(i) It is less ductile than copper
(ii) It is harder than tin
(iii) It can be soldered
(iv) It can be welded

(f) The drawback of electrical seasoning of timber is
(i) Checks
(ii) Splitting
(iii) Cracks
(iv) Reduced strength

(g) Gap grading is one
(i) in which the particles fall within a narrow limit of size fractions
(ii) In which one or more intermediate fractions are absent
(iii) Which combines different fractions of fine and coarse aggregates
(iv) In which all the particles are of uniform size

(h) Which of the following class of engineering ceramics generally included lubricant materials?
(i) Metalloids
(ii) Intermetallics
(iii) Sulphides
(iv) Carbides

(i) Which of the following is not a method of fabrication of rubber?
(i) Vacuum forming
(ii) Buffing
(iii) Calendering
(iv) Compression molding



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(j) Polymer materials become brittle below

(i) Critical temperature (iii) Melting point temperature

(ii) Glass transition temperature (iv) Room temperature

*Q.2* (a) What is meant by the term refractories? Mention the desired properties of refractories. Also state the properties and uses of acidic, basic and neutral refractories. [7]

(b) Explain the tests to be conducted to check the suitability for the construction work. [7]

*Q.3* (a) Enumerate the laboratory tests for cement and describe any two of them. [7]

(b) What are the objectives of preservation of timber? Describe the preservatives which are commonly used in the process of preservation of timber. [7]

*Q.4* (a) Describe manufacturing process of pig-iron by blast furnace. [7]

(b) Enumerate various markets form of steel with neat sketches. [7]

*Q.5* (a) State the properties of aluminium and also discuss the importance of aluminium as structural material. [7]

(b) How is coloured glass made? State the colouring substances which are used for getting different shades of colour of glass. [7]

[7]

*Q.6* (a) Explain briefly various testing procedure for reinforcing steel bars. [7]

(b) What is fracture mechanics. How and why fracture toughness tests are performed in laboratory?

*Q.7* (a) The results of sieve analysis of 600 gm oven dried soil sample are shown below: [7]

| Seive No. | Diameter (mm) | Mass of soil retained on sieve (g) |
|---|---|---|
| 4 | 4.75 | 28 |
| 10 | 2 | 42 |
| 20 | 0.85 | 48 |
| 40 | 0.425 | 121 |
| 60 | 0.25 | 211 |
| 100 | 0.15 | 86 |
| 200 | 0.075 | 40 |
| Pan | - | 24 |

Determine the percent finer than each sieve and plot a grain size distribution curve.

(b) Explain factors affecting ductile to brittle transition temperature (DBTT) [7]

*Q.8* (a) Define structural integrity and discuss briefly various procedures for structural integrity assessment with examples. [7]

(b) What is the difference between True stress-strain and Engineering stress-strain curve? Show with neat sketches. Which one is implemented in design codes and why? [7]

*Q.9* Write short notes on ***any four*** of the following:- [$3\frac{1}{2}$ x 4 = 14]

(a) Flash point and fire point of bitumen

(b) Annealing of glass

(c) Geo-textiles

(d) Acoustic material

(e) Plastic deformation of metals

(f) Graphene

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